

Tackling Loss of Plasticity and Catastrophic Forgetting in Task-Agnostic Continual Reinforcement Learning for Autonomous Driving

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Abstract—Deep Reinforcement Learning (DRL) has become a key technology for autonomous-driving decision-making and control, yet large-scale learning across heterogeneous maps and traffic regimes remains vulnerable to loss of plasticity, catastrophic forgetting, and representation interference. These problems have received increasing attention in continual learning in recent years. Inspired by recent continual-learning studies, we propose a *Similarity-Based Activation and Uncertainty-Aware Stabilization* (SBA-UAS) architecture tailored for reinforcement learning, aiming to endow DRL agents with brain-inspired continual-learning capability. To address loss of plasticity, the architecture uses a similarity-based activation mechanism to dynamically allocate sparse and selectively overlapping neural pathways according to feature similarity in environmental observations, thereby coordinating parameter isolation and feature reuse at the architectural level. To alleviate catastrophic forgetting, SBA-UAS introduces an uncertainty-aware stabilization mechanism that uses Bayesian uncertainty to construct a dynamic Reference Network and combines it with reward-parameter correlation to adaptively regularize critical historical knowledge. Closed-loop simulations under both standard multi-map sampling and diagnostic sequential CARLA streams show that SBA-UAS improves average performance while reducing forgetting relative to the considered baselines. The experiments further suggest that improving Critic-side representation isolation and memory stability can guide an unmodified Actor toward a stable policy. The proposed framework provides a structured approach for strengthening DRL under heterogeneous autonomous-driving training conditions where task-boundary cues are unavailable.

Index Terms—Deep reinforcement learning, continual learning, autonomous driving, loss of plasticity, catastrophic forgetting, bayesian neural networks.

I. INTRODUCTION

REINFORCEMENT learning (RL) has shown great potential for improving traffic efficiency and safety in autonomous driving [1]. However, compared with the strong continual-learning capability of human intelligence, which integrates historical experience in dynamic environments to accelerate the acquisition of new tasks while resisting forgetting [2], existing Deep Reinforcement Learning (DRL) algorithms still face substantial limitations. Most current studies focus mainly on isolated scenarios [3]–[5]. When applied to large-scale, multi-scenario autonomous-driving tasks, a policy

must absorb experience from environments involving heterogeneous maps, route structures, traffic densities, and interaction patterns. Because RL is sample-inefficient, the policy usually requires a long sequence of iterative updates before effective behavior emerges, and this process can easily weaken the plasticity of the underlying neural network for absorbing new information. Moreover, heterogeneous state distributions further induce representation interference and cause previously effective value structures to suffer catastrophic forgetting. This dual dilemma of stability and plasticity is precisely a core challenge long studied in continual learning, and it severely constrains the transfer of autonomous-driving technologies from laboratory environments to complex real traffic systems.

Parameter isolation is an effective structural strategy for mitigating the stability-plasticity dilemma. By assigning different experience domains to partially independent parameter subspaces, this strategy can substantially reduce destructive gradient interference while preserving model capacity for subsequent task adaptation [6]. However, conventional parameter-isolation methods usually rely on explicit task boundaries or task IDs to trigger structural switching, making them difficult to apply directly to continual reinforcement learning environments without clear task partitions. In this paper, *Task-Agnostic* means that the agent receives no explicit task ID or switching signal during training, although this does not imply coverage of all fundamentally different driving regimes. To implement representation-level parameter isolation under this constraint, we propose a *Similarity-Based Activation Network* (SAN). SAN abandons the assumption of hard uniqueness-based scene partitioning; instead, it estimates feature similarity from high-dimensional environmental observations and dynamically activates sparse, partially overlapping neural pathways within the Critic network. This soft partition enables similar scenes to reuse shared features while allocating relatively independent network capacity to more distinct environmental domains, thereby converting the abstract stability-plasticity tradeoff into a data-driven pathway-allocation problem.

On the other hand, regularization-based methods consolidate memory by constraining updates to parameters that are critical for existing knowledge [7], [8]. In DRL, however, such methods are theoretically fragile because value learning depends heavily on bootstrapping targets rather than supervised labels. Importance scores derived from gradient residuals often mix Critic noise, target-network delay, or reward-scale effects, and therefore cannot accurately map stable behavioral knowledge. To address this problem, this paper proposes an

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Uncertainty-Aware Stabilization mechanism. By introducing a Bayesian Neural Network (BNN) as an environment model, the method provides posterior-predictive familiarity signals, so that memory construction no longer depends solely on the value residuals to be stabilized. Low-surprise familiar transitions are stored in a dedicated buffer and used to construct a dynamic Reference Network, thereby providing a continuously updated historical anchor for policy learning without explicit task boundaries. In addition, this paper introduces the *Reward-Parameter Correlation* metric, which defines parameter importance as the accumulated association between parameter updates and batch-level reward variations. This design keeps the regularization objective highly consistent with the reinforcement-learning objective and avoids the limitation of simply equating synaptic value with pure gradient residuals.

The effectiveness of these mechanisms is verified under two experimental settings. First, under the standard multi-map joint-sampling protocol, SBA-UAS significantly improves the agent’s average driving performance across multiple heterogeneous CARLA maps, demonstrating that the continual-learning mechanism effectively enhances policy generalization in large-scale multi-scenario settings. Second, this paper introduces a continual data-stream sampling protocol that switches maps sequentially and no longer generates trajectories from old maps after each switch, using it as a diagnostic stress test to deeply expose loss of plasticity and catastrophic forgetting under long-horizon, non-stationary training. The experiments show that SBA-UAS achieves better performance retention and continual adaptation than existing baselines through similarity-based pathway allocation and uncertainty-aware stabilization. More importantly, the continual-learning enhancement acts only on the action-value network (Critic), without imposing any structural modification on the policy network (Actor). This asymmetric design is rooted in the intrinsic dependency of the Actor-Critic framework: Actor updates are fully governed by Critic value estimation and its gradients. We verify the following view: under sequential domain shifts, as long as the Critic maintains stability between new-environment adaptation and historical value representation, the high-quality value gradients it provides are sufficient to guide the unmodified Actor network toward a globally stable driving policy.

In summary, SBA-UAS constructs a Dual-Memory continual-learning framework at the computational-mechanism level, enabling the agent to efficiently absorb new scenario knowledge in continual autonomous-driving tasks while maintaining historical policy stability. The main contributions of this paper are as follows:

- We propose SBA-UAS, a continual-learning architecture for reinforcement learning. This architecture deeply integrates Similarity-Based Activation with Uncertainty-Aware Stabilization, forming synergy between representation isolation and parameter protection and effectively overcoming the adaptability bottleneck of conventional DRL in sequential, non-stationary CARLA scenarios.
- We construct a **task-agnostic** forgetting-resistance mechanism. By using BNN-based predictive uncertainty estimation to identify and cache “familiar experiences” for building a Reference Network, and combining it with

an adaptive regularization strategy based on Reward-Parameter Correlation, the mechanism addresses the difficulty of accurately evaluating parameter importance without task-boundary cues.

- We reveal a Critic-mediated continual policy evolution mechanism. Experiments verify that, under the Actor-Critic framework, targeted enhancement of Critic memory retention and representation capability, including knowledge reuse, representation isolation, and forgetting suppression, is sufficient to drive an Actor network without structural modification to converge to a stable driving policy under continual domain transfer.

II. RELATED WORK

In the domain of continual learning, researchers have extensively drawn inspiration from neuroscientific mechanisms to mitigate catastrophic forgetting [9]–[12]. Particular focus has been placed on mimicking *synaptic consolidation*, a biological process wherein the brain stabilizes neural connections relevant to past experiences to ensure long-term retention. Building upon this foundation, Synaptic Intelligence (SI) [7] and Elastic Weight Consolidation (EWC) [8] have emerged as representative regularization-based strategies. EWC constrains the magnitude of parameter updates by estimating the Fisher Information Matrix to quantify the influence of each parameter on the network output, whereas SI dynamically modulates weight consolidation based on the path integral of gradient contributions.

However, the applicability of these methods to DRL faces specific challenges. While they are well established in supervised learning for many Artificial Neural Networks (ANNs), their assumptions do not directly transfer to DRL. The primary impediment lies in the nature of DRL optimization: unlike supervised learning, which uses ground-truth labels, DRL relies on estimated gradients derived from bootstrapping targets (e.g., TD error). These estimates contain variance and bias, so calculated parameter-importance metrics can be inaccurate or misaligned with the true policy distribution. Furthermore, beyond gradient-estimation issues, these methods struggle to scale to large-scale continuous scenarios, where performance can degrade as task complexity accumulates.

Alongside catastrophic forgetting, the *loss of plasticity*—the degradation of a model’s ability to acquire new knowledge over time—remains a central concern in continual learning. Several studies address this by constructing multi-model architectures, such as training independent sub-models for individual tasks and integrating their predictions [13]–[15]. Although these approaches isolate inter-task interference, they incur storage overheads that scale linearly with the number of tasks, while learning efficiency is often constrained by the accumulation of model complexity. Masse et al. proposed Context-Dependent Gating (XdG) [6], combining synaptic intelligence with gating mechanisms to learn long task sequences by incorporating task context information. Similarly, Meng et al. [16] leveraged Large Language Models (LLMs) to enhance continual learning in reinforcement learning. However, a limitation shared by these methods is their reliance on explicit

task identifier sequences. Consequently, they are restricted to discrete, distinguishable task scenarios and are ill-suited for continuous, boundary-free tasks, such as policy optimization for autonomous driving in dynamic environments.

To address the demand for general-purpose continual learning without task labels, researchers have introduced alternative approaches. Dohare et al. [17] investigated the phenomenon where deep learning models gradually lose plasticity during continual learning and designed a plasticity injection mechanism to maintain forward transfer capability. However, while this enhances knowledge absorption efficiency, its robustness against catastrophic forgetting remains insufficient. Furthermore, numerous existing methods draw inspiration from the dual-memory mechanism of the cortico-hippocampal circuit [18]–[20]; yet, few studies have explored these principles within the reinforcement learning framework. For instance, CH-HNN [20] employs Spiking Neural Networks (SNNs) to simulate memory consolidation and generalization, achieving task-free continual learning. Nevertheless, SNNs involve high training complexity, and their advantages in low-power consumption are constrained in DRL environments that rely heavily on gradient estimation.

Cross-domain scenario transfer in autonomous driving can be formulated as a continual learning problem characterized by continuous, inseparable environmental domains. The SBA-UAS method proposed in this paper studies policy learning over complex state spaces without relying on task identifiers, and the empirical validation focuses on continual adaptation across CARLA maps with different route structures and traffic conditions. This scope is narrower than full generalization across all heterogeneous driving tasks, but it directly targets the practical setting in which task-boundary cues are unavailable. The central methodological question is therefore how to coordinate representation separation, knowledge reuse, and parameter consolidation when neither task identity nor task-switch timing is observable.

III. PREPARATION

A. Deep Reinforcement Learning and Continual Learning

In DRL, an agent learns to make decisions by interacting with an environment to maximize the long-term cumulative reward. The decision-making problem in autonomous driving can be formalized as a Markov Decision Process (MDP) [21], defined as a tuple (S, A, P, r, γ) , where: S denotes the set of all possible states; A represents the set of all actions the agent can execute; $P : S \times A \times S \rightarrow [0, 1]$ is the state transition probability function, indicating the probability of transitioning to state s' after taking action a in state s ; $r : S \rightarrow \mathbb{R}$ is the reward function; and $\gamma \in (0, 1]$ is the discount factor, which determines the present value of future rewards. The agent's objective is to learn a policy $\pi : S \times A \rightarrow [0, 1]$, which defines a probability distribution over actions given a state, so as to maximize the expected discounted return: $\mathbb{E}_\tau [\sum_{t=0}^{\infty} \gamma^t r(s_t)]$, where $\tau = (s_0, a_0, s_1, a_1, \dots)$ denotes a trajectory, i.e., a sequence of states and actions.

In the formal definition of continual learning, let the task sequence be $\{T_1, T_2, \dots, T_T\}$ and the model parameters



Fig. 1. The agent learns the map scenes sequentially. When the map switches, the previous map is no longer used for new trajectory sampling, although historical transitions from that map may still remain in the replay buffers.

be θ . The goal of continual learning can be characterized by forward transfer and catastrophic forgetting suppression. Forward transfer refers to leveraging past task experience $D_{1:k-1}$ to accelerate learning on the current task T_k , with the optimization objective:

$$\min_{\theta} \mathcal{L}_k(\theta; D_k) \quad \text{subject to} \quad \theta = \arg \min_{\theta'} \sum_{i=1}^{k-1} \mathcal{L}_i(\theta'; D_i). \quad (1)$$

Catastrophic forgetting suppression refers to maintaining performance on previous tasks $\{T_1, T_2, \dots, T_{k-1}\}$ while updating θ to adapt to T_k :

$$\forall i < k, \quad \mathcal{L}_i(\theta_{\text{new}}; D_i) \approx \mathcal{L}_i(\theta_{\text{old}}; D_i). \quad (2)$$

Depending on task boundaries and data distribution characteristics, continual learning is categorized into three types: 1) Task-Incremental Continual Learning (Task-CL), where explicit task identifiers are available and class spaces are mutually exclusive; 2) Domain-Incremental Continual Learning (Domain-CL), where the task objective remains unchanged but the state distribution $P(s)$ shifts, e.g., due to changes in environmental dynamics; and 3) Class-Incremental Continual Learning (Class-CL), where the class space expands without task identifiers.

To effectively study catastrophic forgetting and policy-plasticity degradation in large-scale autonomous-driving decision and control, this paper organizes the training environments as a sequential data stream composed of multiple maps. As shown in Fig. 1, let the map sequence be $\{\text{Map}_1, \text{Map}_2, \dots, \text{Map}_K\}$, where different maps contain both partially shared scenarios and map-specific complex scenarios. This setting constructs a domain-incremental continual learning data stream: the task objective remains unchanged, namely maximizing the expected return in the autonomous-driving decision process; however, the state distribution, scenario composition, traffic-interaction pattern, and environmen-

tal dynamics vary as the map changes. Therefore, the agent must not only adapt to the new map but also preserve policy competence on historical maps. This data stream differs from standard RL data collection because standard RL usually continues to access the training environment according to the target task distribution; it also differs from curriculum learning because curriculum learning may revisit or mix earlier easy data as an optimization aid [31], [32].

It should be emphasized that the sequential map data stream constructed in this paper is used only to validate the continual-learning capability of the algorithm in the reinforcement-learning setting and is not a necessary component of SBA-UAS itself. In normal use of SBA-UAS, training data should be collected by the standard off-policy protocol according to the target environment distribution, for example, by jointly sampling the available training maps or by mixed sampling according to target-scenario weights. The sequential data stream is only an experimental protocol, used to more clearly evaluate the method's forward adaptation ability, historical-performance retention ability, and catastrophic-forgetting suppression ability when the environment distribution changes continually.

B. MDP Formulation

1) *State Space*: This paper focuses on autonomous driving decision and control tasks using Bird's Eye View (BEV) images as state input, aiming to maintain compatibility with a broader range of end-to-end approaches [22], [23]. Inspired by Roach [22], we adopt BEV semantic segmentation images $o \in [0, 1]^{W \times H \times C}$ to reduce task complexity. In our setup, the number of channels $C = 15$, including: drivable area layer, desired route layer, lane boundary layer; four stacked ego-vehicle state layers; four pedestrian position segmentation layers; and four traffic signal and sign layers (including traffic light stop lines, stop trigger zones, etc.). The stacked channel design helps capture motion states and dynamic information of traffic participants. Additionally, a measurement vector $m \in \mathbb{R}^6$ is introduced to supplement ego-vehicle state information not fully captured in the BEV image, including ground-truth measurements of steering, throttle, brake, gear, lateral velocity, and longitudinal velocity. Fig. 2 illustrates the rendered BEV segmentation image after stacking.

2) *Action Space*: In the SBA-UAS implementation, the action space is defined as $a \in [-1, 1]^2$. The longitudinal component a_1 is treated as a normalized desired-speed intention rather than an absolute speed command. During training, a_1 is affinely mapped to $v_{\text{tar}} \in [0, 6]$ m/s for all maps, with $a_1 < -0.7$ forcing $v_{\text{tar}} = 0$; highway-like scenes additionally attenuate available braking authority to approximate longer stopping-distance constraints within this shared scale. The target speed is converted into longitudinal commands by a PID controller, while a_2 is mapped to a normalized steering angle and smoothed by a low-pass filter. For speed-scaled evaluation, we keep the policy fixed and replace the upper bound in the affine speed mapping with the desired road speed, with the PID layer constraining throttle, brake, and steering to dynamically feasible responses. To model bounded continuous actions, we

employ the Beta distribution $\mathcal{B}(\alpha, \beta)$ over $[0, 1]$ and affinely transform samples to the required interval [33]; its bounded support and flexible shape keep sampled intentions within the admissible control interface.

3) *Reward Design*: To guide the agent in learning a stable policy that balances safety, compliance, and traffic efficiency in complex traffic environments, we designed a composite reward function comprising sparse terminal penalties and dense behavioral shaping rewards. The task terminates and triggers a termination reward r_{terminal} when the agent meets any of the following conditions:

- Violating traffic rules (running a red light/stop sign), $r_{\text{terminal}} = -1 - v$ (v is the ego-vehicle speed);
- Physical collision, $r_{\text{terminal}} = -1 - v$;
- BEV boundary box overlap detection (e.g., obstacle collision), $r_{\text{terminal}} = -1 - v$;
- Route deviation (lateral distance > 3.5 m), $r_{\text{terminal}} = -1$;
- Vehicle stagnation (speed < 0.1 m/s for 120 s), $r_{\text{terminal}} = -1$.

The reward function guides the agent to optimize decision-making behaviors through multi-dimensional metrics, specifically designed as follows:

- Speed reward: $r_{\text{speed}} = 1 - \frac{|v - v_{\text{desired}}|}{v_{\text{max}}}$, where v is the real-time speed, $v_{\text{max}} = 6$ m/s is the shared training speed cap; $v_{\text{desired}} = v_{\text{max}}$ when there are no obstacles on the desired route, otherwise it linearly decreases to 0 based on the distance between the obstacle and the ego-vehicle.
- Position reward: $r_{\text{position}} = -0.5\Delta_p$, where Δ_p is the lateral distance (in meters) between the center of the ego-vehicle and the centerline of the target route, used to constrain path tracking accuracy.
- Orientation reward: $r_{\text{rotation}} = -\Delta_r$, where Δ_r is the absolute difference (in radians) between the heading angle of the ego-vehicle and the heading angle of the target route, used to align the driving direction.
- Action smoothness reward: $r_{\text{action}} = -0.1$ if $|\text{steering}_t - \text{steering}_{t-1}| > 0.01$ else 0, penalizing frequent steering adjustments to reduce policy jitter and improve control stability.

To prevent the agent from getting stuck in local optima (e.g., undesirable behaviors such as remaining stationary for extended periods), we employ a dynamic penalty coefficient strategy [24]. Specifically, during the early stages of training, an initially minimal penalty coefficient w_p is introduced on r_{rotation} and r_{action} . This coefficient is then gradually increased at a fixed step until it reaches 1, thereby increasing the penalty intensity for inefficient behaviors over the course of training and guiding the policy toward more feasible solutions.

IV. METHODOLOGY

Fig. 2 depicts the overall workflow of the SBA-UAS architecture. By synergistically optimizing *Representation Isolation* and *Parameter Stabilization*, this framework endows the agent with the capacity for continual adaptation in task-agnostic CARLA environment streams. The system is primarily composed of two coupled subsystems: the Deep Reinforcement

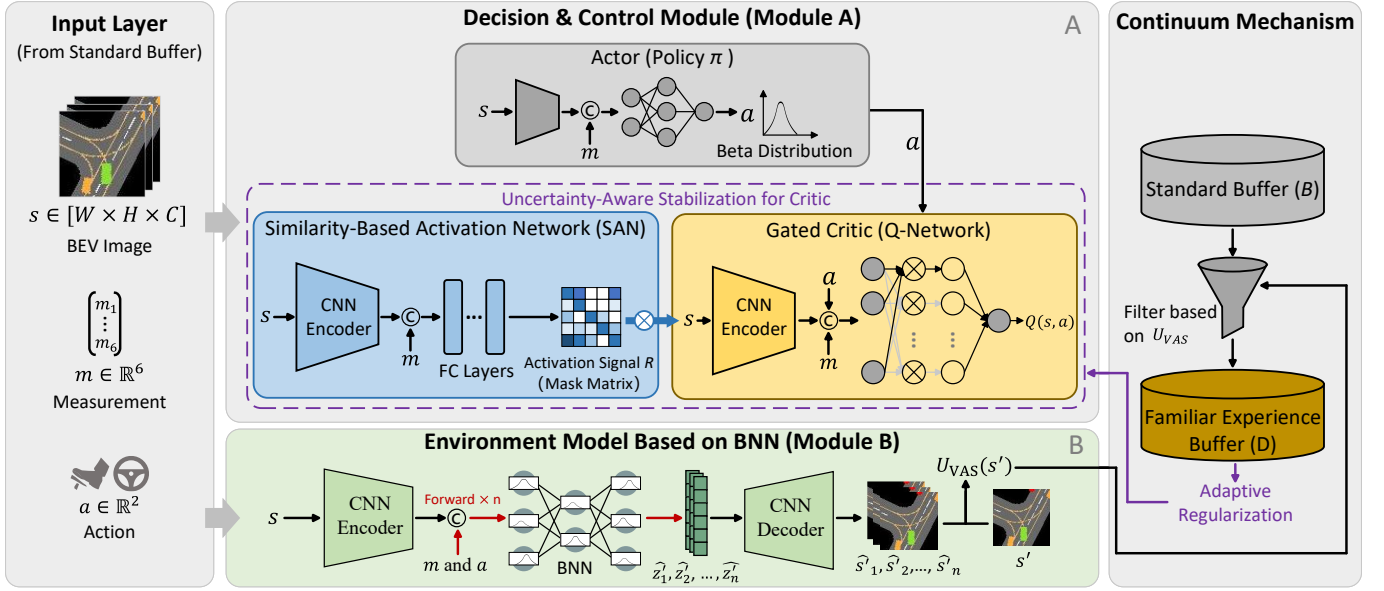


Fig. 2. Schematic illustration of the *Similarity-Based Activation and Uncertainty-Aware Stabilization* (SBA-UAS) framework. The architecture comprises two synergistically coupled components: the Reinforcement Learning Module (Module A) and the Environment Modeling Module (Module B). **Module A** encompasses two parallel pathways: the upper stream features a *Standard Actor*, which employs a structurally unmodified conventional policy network to map environmental states directly to action policies; the lower stream constitutes the Critic component, which integrates a *Similarity-Based Activation Network* (SAN) to generate gating signals that dynamically modulate the Q-network. This mechanism facilitates the effective reuse of task-relevant knowledge and the disentanglement of representations during the value estimation process. **Module B** implements an *Uncertainty-Aware Stabilization* mechanism. It leverages an environment model to estimate state-dependent uncertainty, governing the selective storage of “familiar” experiences. Building upon these buffered experiences, it imposes adaptive regularization on the Critic’s parameters, effectively mitigating catastrophic forgetting within the value function and thereby stabilizing the evolutionary trajectory of the Actor’s policy.

Learning backbone (Module A) and the Probabilistic Environment Modeling Unit (Module B). In Subsection A, we detail Module A, explicitly articulating the gating activation mechanism and update strategy of SAN to resolve the challenge of isolating varying scenario features while simultaneously reusing general knowledge. Subsequently, Subsection B introduces Module B, which details the design of a probabilistic environment model based on BNN, aimed at establishing a reliable uncertainty metric benchmark for continual learning. Finally, Subsection C elucidates how this uncertainty metric is leveraged to construct a “Familiar Experience Buffer” and, in conjunction with the Reward-Parameter Correlation metric, to implement adaptive regularization, thereby mitigating catastrophic forgetting in non-stationary dynamic environments.

A. SAN and Gated Pathway Activation in the Critic

The Similarity-Based Activation Network (SAN) is responsible for abstracting scene features and integrating prior knowledge to modulate the activation of corresponding pathways in the Q-value network. Its dynamic pathway-allocation principle is adapted from cortico-hippocampal continual learning architectures [20], but is reformulated here for Critic-side modulation in DRL. As illustrated in Fig. 2(A), SAN first encodes the BEV input using a convolutional neural network (CNN), and then passes the encoded representation through a mapping module composed of two shared fully connected layers and n parallel linear projection heads. Let $\mathbf{Z} = [\mathbf{z}_1; \dots; \mathbf{z}_n] \in \mathbb{R}^{n \times c}$ denote the resulting gate-logit ma-

trix. To avoid constraining each row to a probability simplex, SAN does not apply a row-wise Softmax over the c neurons. Instead, the activation signal is computed by an element-wise sigmoid gate, $\mathbf{R} = \sigma(\mathbf{Z})$, $\mathbf{Z} = G(s, m; \theta_s)$, where θ_s denotes the parameters of SAN. Here, $\mathbf{R} \in (0, 1)^{n \times c}$, with c equal to the number of neurons in the hidden layers of the Q-network to be modulated, and n representing the number of such layers; the i -th row is denoted as $\mathbf{R}_i$. Thus, $\mathbf{R}_i$ is a bounded soft gating vector for the i -th hidden layer rather than a probability distribution whose entries must sum to one. To enable the context encoder to output a matrix of this structure, the final stage of SAN employs six parallel linear projection heads, each containing 2048 neurons. Consequently, in this work, $n = 6$ and $c = 2048$. Additionally, the first and second shared layers of the mapping module each contain 1024 neurons.

The SAN is trained using the following loss function:

$$\mathcal{L}_{\text{SAN}}(\theta_s) = \mathbb{E}_{s, \tilde{s}} \sum_{i=1}^n \left[\left| \text{sim}(\mathbf{R}_i, \tilde{\mathbf{R}}_i) - \text{sim}(s, \tilde{s}) \right|^2 + \beta \left| \|\mathbf{R}_i\|_1 - \rho c \right| \right], \quad (3)$$

where s and $\tilde{s}$ are two distinct states sampled from different experiences stored in the DRL replay buffer. The function $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, computed based on the CNN-encoded feature vectors rather than the raw BEV inputs. The first term enforces consistency between the similarity of activation patterns $\mathbf{R}_i$ and $\tilde{\mathbf{R}}_i$ and the similarity of their corresponding input states s and $\tilde{s}$. The second term imposes

an activation-budget constraint on each gating row: since $\mathbf{R}_i$ is produced by element-wise sigmoid gates, $\|\mathbf{R}_i\|_1$ is not fixed and approximates the soft number of active neurons in the i -th Q-network layer. The parameter β balances the two objectives, while ρ denotes the target active-neuron ratio, so that ρc is the desired soft activation budget per modulated layer. This term therefore yields non-zero gradients whenever the total gate mass deviates from the target budget and directly encourages sparse pathway usage. This objective follows the activation-similarity idea in [20], while the BEV feature encoding and Critic modulation are specific to our DRL setting.

To align with the matrix structure of the neuron activation signal $\mathbf{R}$, the backbone of our Q-value network is designed as a fully connected network with n layers, each containing c neurons. Specifically, the hidden state at layer k is computed as:

$$\mathbf{h}_k = \sigma(\mathbf{W}_k^\top \mathbf{h}_{k-1} + \mathbf{b}_k) \odot \mathbf{R}_k, \quad k = 1, 2, \dots, n, \quad (4)$$

where $\mathbf{W}_k \in \mathbb{R}^{c \times c}$ for $k \neq 0, n$, $\mathbf{h}_k \in \mathbb{R}^c$, and $\odot$ denotes element-wise multiplication. The forward pass of the Q-network is denoted as $\hat{q}_{(s,a)} = Q(s, m, a; \phi, \mathbf{R}_s)$, where ϕ represents the parameters of the Q-network. This gated modulation is adapted from the pathway-isolation mechanism in [20]. When the encoded state similarity is high, similar pathways in the Q-network are activated; conversely, dissimilar scene observations trigger largely distinct pathways. Moreover, certain pathways in the Q-network are reusable, enabling the sharing of representations for common features across states.

The Q-network is trained to maximize the expected discounted return in DRL and is updated via bootstrapping using the Mean Squared Bellman Error (MSBE):

$$\begin{aligned} \mathcal{L}_Q(\varphi_k) = & \mathbb{E}_{\substack{(s_t, m_t, a_t, r_t, s_{t+1}, m_{t+1}) \sim \tau \\ \hat{a}_{t+1} \sim \pi(\cdot | s_{t+1}, m_{t+1}; \psi)}} \left[Q(s_t, m_t, a_t; \varphi_k, \mathbf{R}_{s_t}) \right. \\ & \left. - \left(r_t + \gamma \min_{k=1,2} Q(s_{t+1}, m_{t+1}, \hat{a}_{t+1}; \varphi_k, \mathbf{R}_{s_{t+1}}) \right) \right]^2, \end{aligned} \quad (5)$$

where $\pi(\cdot | s_{t+1}, m_{t+1}; \psi)$ denotes the policy network parameterized by ψ . We employ the Double Q-learning technique to mitigate overestimation bias. The policy is optimized via the following loss:

$$\begin{aligned} L_\pi(\psi) = & \mathbb{E}_\tau \left[\mathbb{E}_{\hat{a}_t \sim \pi} \left[- \min_{k=1,2} Q(s_t, m_t, \hat{a}_t; \varphi_k, \mathbf{R}_{s_t}) \right. \right. \\ & \left. \left. + \alpha \log \pi(\hat{a}_t | s_t, m_t; \psi, \mathbf{R}_{s_t}) \right] \right], \end{aligned} \quad (6)$$

where the entropy regularization term $\log \pi(\hat{a}_t | s_t, m_t; \psi)$ enhances exploration, and α is a temperature hyperparameter.

The aforementioned methodology realizes both the isolation of learning across varying CARLA driving scenarios and the reuse of transferable knowledge. Prior work [20] supports the role of such architectures in continual learning, while Masse et al. [6] show that this “isolation” strategy alone is insufficient to prevent catastrophic forgetting. The system therefore also requires parameter stabilization mechanisms—such as Synaptic Intelligence or Elastic Weight Consolidation—to protect critical weights from being overwritten.

However, existing parameter stabilization mechanisms encounter two obstacles when directly applied to DRL. First, DRL training distributions are non-stationary, and gradient estimates contain noise, which makes parameter importance difficult to quantify. Second, most of these approaches rely on explicit task identifiers to demarcate “old task” boundaries—information that is unavailable in the task-agnostic CARLA environment stream considered here. To address these limitations, we propose an *Uncertainty-Aware Stabilization* mechanism that is compatible with the DRL training paradigm and operates without task context cues. Specifically, in the subsequent Subsection B, we introduce a BNN-based environment model to compute the shifted VAS score $\tilde{U}_{\text{VAS}}$ —a metric serving as a proxy for “scene familiarity” whose theoretical basis is formalized in Section IV-B. Subsequently, in Subsection C, we leverage $\tilde{U}_{\text{VAS}}$ to filter high-confidence historical data for the construction of a “Familiar Experience Buffer.” Building upon these familiar experiences, we design an adaptive regularization technique that identifies and consolidates critical network parameters in an unsupervised manner.

B. Bayesian Environment Model for Uncertainty Estimation

To identify “familiar experiences” in the absence of explicit task boundaries, we construct a probabilistic environment model based on BNN. In Bayesian predictive modeling, posterior variation expresses how strongly the learned dynamics can explain an observed transition under sampled model hypotheses [25]. We use this principle to derive a familiarity criterion that is tied to latent transition predictability rather than to the current Critic residual alone.

By modeling network parameters as probability distributions rather than deterministic values, BNN explicitly represent model uncertainty [25]. Let θ_B denote the parameters of the BNN, which can be viewed as an instantiation of the environment model. We place an isotropic Gaussian prior over the parameters: $p(\theta_B) = \prod_{i=1}^K \mathcal{N}(\theta_{B,i}; 0, \sigma_i^2)$. The posterior $p(\theta_B | s, a)$ is approximated by a factorized Gaussian variational distribution:

$$\begin{aligned} q(\theta_B | \phi) = & \prod_{i=1}^K \mathcal{N}(\theta_{B,i}; \mu_i, \sigma_i^2), \\ \text{s.t. } \phi = & \{(\mu_1, \sigma_1), \dots, (\mu_K, \sigma_K)\}, \end{aligned} \quad (7)$$

where K is the total number of parameters.

As illustrated in Fig. 2(B), the environment model adopts an autoencoder-like architecture. We employ a ResNet-18 backbone [26] as both the encoder and decoder networks, with the BEV observation s represented at $128 \times 128 \times 5$. The forward pass of the environment model is defined as $z = \text{ENC}(s; \theta_E)$, $\hat{z}' = \text{BNN}(z, m, a; \theta_B)$, and $\hat{s}' = \text{DEC}(\hat{z}'; \theta_D)$ (8), where θ_E , θ_D , and θ_B denote the parameters of the encoder, decoder, and BNN, respectively, and $z, \hat{z}' \in \mathbb{R}^{256}$. To account for both reconstruction fidelity and predictive accuracy, we define the

following composite loss function:

$$\begin{aligned} \mathcal{L}_{\text{ENV}}(\theta_E, \theta_B, \theta_D) = & \mathbb{E}_{(s, m, a, s') \sim \tau} \left[\|s - \text{DEC}(z; \theta_D)\|^2 \right. \\ & + \|s' - \text{DEC}(\text{ENC}(s'; \theta_E); \theta_D)\|^2 \\ & + \|\text{ENC}(s'; \theta_E) - \text{BNN}(z, m, a; \theta_B)\|^2 + \sum_{i=1}^n \|s' - \hat{s}'_i\|^2 \\ & \left. - D_{\text{KL}}[q(\theta_B | \phi) \| p(\theta_B)] \right], \end{aligned} \quad (9)$$

where the first two terms are standard autoencoding reconstruction losses, the third aligns the BNN’s latent prediction with the encoded next state, the fourth term penalizes prediction error in observation space, and the final term maximizes the Evidence Lower Bound (ELBO) by regularizing the variational posterior toward the prior.

To assess the agent’s “familiarity” with the current transition, we follow *Variational Assorted Surprise Exploration* (VASE) [27], which uses a variational BNN dynamics model to evaluate predictive surprise. Here, the VAS score is used as a replay-filtering criterion, not as VASE’s full intrinsic reward or a pure epistemic-uncertainty decomposition. The expected negative log-likelihood is

$$U_{\text{VAS}}(s, a, s') = \mathbb{E}_{\theta_B \sim q(\theta_B | \phi)} [-\log p(s' | s, a; \theta_B)]. \quad (10)$$

For the normalized next-state vector $s' \in \mathbb{R}^d$, we assume an isotropic Gaussian observation model, $p(s' | s, a; \theta_B) = (2\pi\sigma_c^2)^{-d/2} \exp(-\|s' - s\|^2 / (2\sigma_c^2))$. Since its normalization term does not affect sample ranking, the operational shifted score is

$$\tilde{U}_{\text{VAS}}(s, a, s') = \mathbb{E}_{\theta_B \sim q(\theta_B | \phi)} \left[\frac{\|s' - s\|^2}{2\sigma_c^2} \right]. \quad (11)$$

The expectation in Eq. (11) is approximated via Monte Carlo sampling:

$$\tilde{U}_{\text{VAS}}(s, a, s') \approx \frac{1}{N} \sum_{n=1}^N \frac{\|\hat{s}'_n - s'\|^2}{2\sigma_c^2}, \quad (12)$$

where $\theta_{B,n} \sim q(\theta_B | \phi)$. With $z = \text{ENC}(s; \theta_E)$, we perform $N = 64$ stochastic forward passes and decode $\hat{s}'_n = \text{DEC}(\text{BNN}(z, m, a; \theta_{B,n}))$.

For well-learned old experiences, the Bayesian dynamics model tends to produce accurate and consistent predictions, resulting in a low shifted VAS score. In contrast, unseen or poorly modeled transitions may produce larger prediction errors or stronger posterior-sample disagreement, thereby increasing $\tilde{U}_{\text{VAS}}$. Thus, the score provides an unsupervised transition-familiarity signal for replay filtering without task labels. It is not a safety-criticality measure or a complete epistemic/aleatoric decomposition, and the environment model is used only for training-time memory construction rather than deployed policy inference.

C. Parameter Stabilization: Adaptive Regularization Mechanism for Mitigating Catastrophic Forgetting

Fig. 3 illustrates how the *Familiar Experience Buffer* $\mathcal{D}$ is constructed during deep reinforcement learning (DRL)

using $\tilde{U}_{\text{VAS}}$. Specifically, SBA-UAS maintains two replay buffers: $\mathcal{B} = \{(s_t, m_t, a_t, r_t, s_{t+1}, d_t, \tilde{U}_{\text{VAS}}(s_t, a_t, s_{t+1}))\}_{t=1}^m$ and $\mathcal{D} = \{(s_t, m_t, a_t, r_t, s_{t+1}, d_t, \tilde{U}_{\text{VAS}}(s_t, a_t, s_{t+1}))\}_{t=1}^{m'}$, where $d_t \in \{0, 1\}$ indicates whether the episode terminates at time t . Newly observed experiences are first annotated with their $\tilde{U}_{\text{VAS}}$ values by the environment model and then stored in buffer $\mathcal{B}$. If $\mathcal{B}$ is full, the incoming experience replaces the entry $o \in \mathcal{B}$ with the *smallest* $\tilde{U}_{\text{VAS}}$ value (i.e., highest familiarity). The evicted entry o is then transferred to buffer $\mathcal{D}$. Similarly, if $\mathcal{D}$ is full, o replaces the experience in $\mathcal{D}$ with the *largest* $\tilde{U}_{\text{VAS}}$ value (i.e., lowest familiarity), and the displaced experience is permanently discarded. Consequently, after an environment switch, both buffers can still preserve historical data from the immediately previous map and even older maps; what disappears is only further trajectory collection from the old active map. In cases where multiple entries share identical extremal $\tilde{U}_{\text{VAS}}$ values, uniform random sampling is used to resolve ties. The rule should be interpreted as a consolidation criterion, not a safety-prioritization policy: because $\tilde{U}_{\text{VAS}}$ measures familiarity rather than event criticality, rare high-uncertainty events may be under-retained. Future extensions should combine uncertainty with risk or event-criticality criteria for safety-sensitive memory selection.

The experiences stored in $\mathcal{D}$ represent frequently encountered scenarios—referred to as *old memory*—in which the agent typically exhibits stable and high-performing policies. In the following, we describe how these familiar experiences are leveraged to consolidate the agent’s long-term knowledge.

Following classical parameter stability methods such as Synaptic Intelligence (SI), Elastic Weight Consolidation (EWC), and XdG-style gated continual learning [6]–[8], a regularization term $\mathcal{L}_{\text{ps}}$ is added to the original loss to protect critical parameters:

$$\mathcal{L}_{\text{ps}}(\varphi) = \eta \sum_i \Omega_i (\varphi_i - \varphi_i^{\text{prev}})^2, \quad (13)$$

where φ_i denotes the i -th parameter of the Q-network, φ_i^{prev} is its value at the end of the previous task, and Ω_i quantifies its importance. However, these approaches rely on explicit task boundaries and prior knowledge of the task sequence—assumptions that do not hold in our continuous, task-agnostic learning setting where task switches are unpredictable.

To address this limitation, we reformulate the mechanism as follows. Within an Actor-Critic architecture, we train the primary Q-network using data from buffer $\mathcal{B}$, denoted as $Q(z, a; \varphi, \mathbf{R}_s, \mathcal{B})$. Simultaneously, we maintain a secondary Q-network—initialized with different random weights and trained exclusively on data from buffer $\mathcal{D}$ —denoted as $Q_F(z, a; \varphi^{\text{prev}}, \mathbf{R}_s, \mathcal{D})$. Here, φ_i^{prev} is redefined as the i -th parameter of this auxiliary “familiarity” network Q_F . Consequently, the Q-network loss in DRL is revised to:

$$\mathcal{L}'_Q(\varphi) = \mathcal{L}_Q(\varphi) + \mathcal{L}_{\text{wp}}(\varphi). \quad (14)$$

We also explored an alternative strategy where ϕ^{prev} is periodically updated to the current network parameters at fixed iteration intervals (e.g., every $e, 2e, \dots$ steps). However, empirical evaluation showed no performance gain, and thus

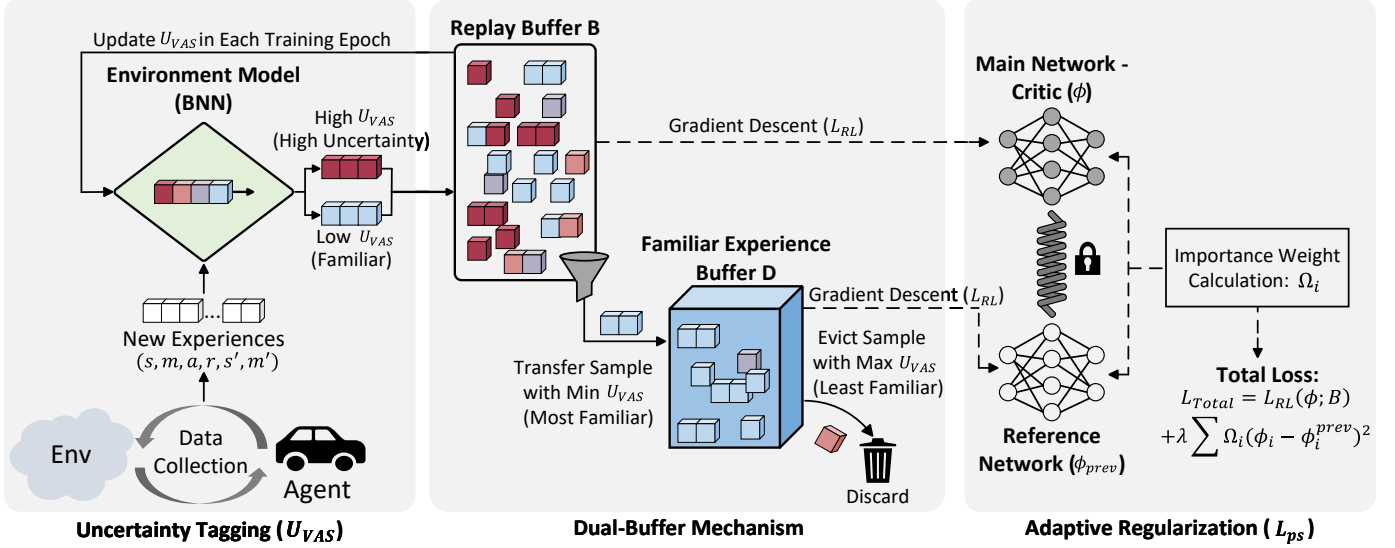


Fig. 3. Schematic illustration of the Familiar Experience Buffer construction and the Adaptive Regularization mechanism. **(Left) Uncertainty Annotation:** Incoming experiences are evaluated by the BNN-based environment model and assigned a shifted VAS score $\tilde{U}_{VAS}$. **(Center) Dual-Buffer Dynamics:** All experiences initially enter the Replay Buffer, B . Upon the saturation of B , the experience with the lowest $\tilde{U}_{VAS}$ value (indicating highest familiarity) is transferred to the Familiar Experience Buffer, D . Conversely, when D reaches capacity, the experience with the highest $\tilde{U}_{VAS}$ value (indicating least familiarity) is evicted. **(Right) Synaptic Protection Mechanism:** Data retained in Buffer D serves as reference representations for historical knowledge. Importance weights (Ω_i) are computed based on parameter sensitivity to impose an adaptive penalty term ($\mathcal{L}_{ps}$) on the parameter updates of both the Critic and the SAN, thereby preventing catastrophic forgetting of previously acquired policies.

this variant was discarded. Analogously, we apply the same parameter stability mechanism to the SAN:

$$\mathcal{L}'_{SAN}(\theta_s) = \mathcal{L}_{SAN}(\theta_s) + \mathcal{L}_{ps}(\theta_s). \quad (15)$$

In this context, $\theta_{s,i}^{prev}$ now denotes the i -th parameter of a separately initialized SAN trained solely on buffer D . This design enables unsupervised, continual consolidation of both policy representations and activation pathways without requiring task identifiers or explicit boundary signals.

Next, we describe how Ω_i in Eq. (13) is computed when applying the weight protection mechanism to the Q-value network in DRL. Inspired by SI/XdG importance accumulation [6], [7], but differing from their loss-gradient-based formulation, we use reward-parameter correlation to better match the DRL objective. In DRL, the policy loss function depends on estimates of future discounted returns. Therefore, directly using parameter gradients derived from the DRL loss function to compute Ω_i would be inaccurate. To address this, we adopt an alternative approach based on the correlation between parameter updates and changes in reward. Specifically, for the mini-batch $\mathcal{M}_t$ used in the t -th training update, we define $\bar{r}(t) = |\mathcal{M}_t|^{-1} \sum_{(s,a,r,s') \in \mathcal{M}_t} r$ as the batch-mean immediate reward. We then calculate the product of the parameter update $\varphi(t) - \varphi(t-1)$ between adjacent training batches and the concurrent change in batch-mean reward $\bar{r}(t) - \bar{r}(t-1)$, accumulating this product over all training batches t :

$$\omega_i = \sum_t (\varphi_i(t) - \varphi_i(t-1))(\bar{r}(t) - \bar{r}(t-1)), \quad (16)$$

where t denotes the index of the training batch. For normalization, we first compute the sum of the absolute values of

these products, which represents the maximum possible value of the above expression:

$$\bar{\omega}_i = \sum_t |\varphi_i(t) - \varphi_i(t-1)| \cdot |\bar{r}(t) - \bar{r}(t-1)|. \quad (17)$$

Then, we obtain the normalized Ω_i :

$$\Omega_i = \left| \frac{\omega_i}{\bar{\omega}_i + \zeta} \right|, \quad (18)$$

where ζ is a damping term, typically set to a small fixed value. For the SAN, since its loss function does not involve bootstrapped value estimation, we compute ω_i using an SI-style gradient contribution [7]:

$$\omega_i = \sum_t (\theta_{s,i}(t) - \theta_{s,i}(t-1)) \left(-\frac{\partial \mathcal{L}_{SAN}(\theta_s)}{\partial \theta_{s,i}} \right). \quad (19)$$

Finally, Ω_i for SAN is obtained as:

$$\Omega_i = \max \left(0, \frac{\omega_i}{\bar{\omega}_i + \zeta} \right). \quad (20)$$

In summary, we have adapted the original Synaptic Intelligence parameter stabilization method to not only apply to continuous lifelong learning tasks without explicit task sequence markers but also to scenarios where gradient estimates are integral, such as in DRL. Algorithm 1 summarizes the pseudocode for the entire SBA-UAS process.

V. EXPERIMENTS AND DISCUSSION

A. Experimental Setup

All evaluations were conducted in the CARLA 0.9.11 simulation environment. We assessed the proposed method on

Algorithm 1 SBA-UAS Algorithm

Require: Map sequence $\mathcal{M}$, Buffers $\mathcal{B}, \mathcal{D}$
Ensure: Policy π_ψ , Critic Q_ϕ , SAN S_{θ_s}

Note: the ordered map loop is only a continual-learning evaluation protocol; standard use follows ordinary off-policy sampling from the target training environments.

- 1: **for** each map $M_k \in \mathcal{M}$ and step t **do**
 - Phase 1: Decision & Uncertainty (Independent Encoders)**
 - 2: Get activation $\mathbf{R}_t \leftarrow S(s_t; \theta_s)$ and action $a_t \sim \pi(\cdot | s_t, \mathbf{R}_t)$
 - 3: Execute $s_{t+1}, r_t \leftarrow M_k.\text{step}(a_t)$
 - 4: Compute U_t via BNN ensemble with isolated encoder E_{θ_E}
 - 5: $e_t \leftarrow (s_t, a_t, r_t, s_{t+1}, U_t)$
 - Phase 2: Dual-Buffer Maintenance**
 - 6: Store e_t in $\mathcal{B}$. If $\mathcal{B}$ full:
 - 7: Move lowest U (familiar) sample $e_{\mathcal{B}}^*$ from $\mathcal{B}$ to $\mathcal{D}$
 - 8: If $\mathcal{D}$ full, remove highest U (novel) sample from $\mathcal{D}$
 - Phase 3: Stabilization & Update**
 - 9: **if** training conditions met **then**
 - 10: Update Env Model Θ_{env} using $\mathcal{B} \cup \mathcal{D}$
 - 11: Update Ref. Networks $\phi_{\text{ref}}, \theta_{s,\text{ref}}$ using $\mathcal{D}$ only
 - 12: Update Importance Weights Ω via Reward Correlation
 - 13: $\mathcal{L}_{\text{reg}} \leftarrow \sum \Omega(\theta - \theta_{\text{ref}})^2$
 - 14: Update Q_ϕ, S_{θ_s} min. $\mathcal{L}_{\text{RL}}(b_{\mathcal{B}}) + \lambda \mathcal{L}_{\text{reg}}$
 - 15: Update π_ψ min. $\mathcal{L}_{\text{Actor}}(b_{\mathcal{B}})$
 - 16: **end if**
 - 17: **end for**
-

the LeaderBoard benchmark [28], which defines a complex driving task spanning six maps and representative traffic scenarios, including four-way intersections (US-style), roundabouts, stop-sign-controlled junctions, lane changes, and merging maneuvers. Traffic density was configured to the “busy” setting, corresponding to 30 vehicles per kilometer. During training, the ego vehicle and social vehicles obeyed the same 6 m/s speed cap across all maps; highway-like routes further used attenuated braking authority, did not introduce additional pedestrian actors, and retained the benchmark default pedestrian/static-obstacle settings. The sequential map order is used as a fixed domain-incremental experimental stream: the six maps contain partially shared scenarios as well as map-specific complex scenarios, and this order should not be interpreted as a universal ranking of CARLA map difficulty or as a curriculum-learning claim.

Metrics. Our evaluation metrics fall into three categories: Route Completion rate (RC), Infraction Score (IS), and the LeaderBoard-defined Driving Score (DS). RC denotes the percentage of route distance successfully completed without collisions or timeouts. DS is defined as the product of the route completion ratio and an infraction penalty factor that consolidates all triggered violations into a unified discount term. DS serves as a fine-grained metric accounting for a broad spectrum of infractions, making it well-suited for evaluating long-horizon driving performance. To assess continual learning

capabilities, we further employ three standard metrics:

- **Average Performance (AP):**

$$\text{AP}(Y) = \frac{1}{T} \sum_{i=1}^T Y_{T,i}, \quad (21)$$

where $Y \in \{\text{RC}, \text{IS}, \text{DS}\}$. When $Y = \text{DS}$, $\text{DS}_{T,i}$ denotes the driving score of the final model evaluated on task i .

- **Average Forgetting (AF):**

$$\text{AF}(Y) = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{k < T} Y_{k,i} - Y_{T,i} \right), \quad (22)$$

which quantifies performance degradation on previously learned tasks; lower values indicate reduced forgetting.

- **Forward Transfer (FWT):**

$$\text{FWT}(Y) = \frac{1}{T-1} \sum_{i=2}^T (Y_{i-1,i} - \bar{b}_i), \quad (23)$$

where $\bar{b}_i$ is the baseline performance of a randomly initialized model on task i . FWT measures the zero-shot benefit that historical knowledge confers upon the initial performance of a new task.

Baselines. For comparative evaluation, we selected several representative baseline methods:

- 1) **PPO (Roach)** [22]: A state-of-the-art DRL agent used in two configurations. In the standard RL agent-sampling reference, PPO (Roach) is trained from data collected jointly across all six maps and evaluated by averaging its scores over the same maps. In the continual-learning comparison, PPO (Roach) follows the same sequential map protocol as the other methods.
- 2) **SAC with large replay buffer:** To evaluate the potential of off-policy RL in mitigating catastrophic forgetting, we implemented Soft Actor-Critic (SAC) [29] equipped with an extensive replay buffer with a capacity equivalent to one-third of the total training steps.
- 3) **CH-HNN** [20]: An advanced continual learning architecture inspired by the cortical-hippocampal circuit, originally designed for task-agnostic vision tasks. Although developed for supervised learning, its core mechanisms—dynamic pathway allocation and memory consolidation—are inherently extensible to DRL. We adapted CH-HNN to our experimental setting for fair comparison.
- 4) **XdG** [6]: A hybrid continual learning method integrating Elastic Weight Consolidation (EWC), Synaptic Intelligence (SI), and context-dependent gating. XdG has been reported to achieve higher performance than individual components (e.g., EWC, SI, or ensemble learning) on MNIST, Tiny-ImageNet, and RL benchmarks. Since XdG requires explicit task identifiers, we provided it with ground-truth map indices during training, corresponding to the sequence in which maps were encountered.

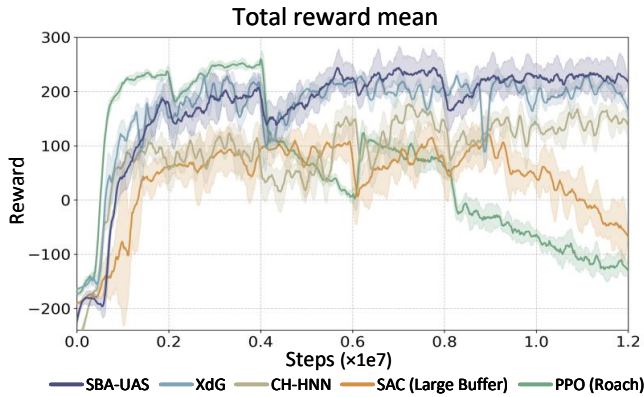


Fig. 4. Average total reward convergence curves of the proposed SBA-UAS method and baseline algorithms across different map domains. Following an initial warm-up phase, each map is trained for 2 000 000 environment steps. All experiments are repeated 10 times with distinct random seeds to ensure statistical robustness. Shaded regions represent one standard deviation around the mean.

TABLE I

AVERAGE PERFORMANCE UNDER THE STANDARD RL AGENT-SAMPLING PROTOCOL. VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER 10 RANDOM SEEDS AND AVERAGED OVER THE SIX EVALUATION MAPS. THE COLUMNS THEREFORE REPORT AP(DS), AP(RC), AND AP(IS). XdG IS EXCLUDED BECAUSE IT RELIES ON EXPLICIT TASK-SEQUENCE IDENTIFIERS.

Method	AP(DS) $\uparrow$	AP(RC) $\uparrow$	AP(IS) $\uparrow$
SBA-UAS	76.85$\pm$4.41	90.16$\pm$3.13	0.846$\pm$0.018
PPO (Roach)	72.45 $\pm$ 3.12	88.26 $\pm$ 1.80	0.81 $\pm$ 0.038
SAC	67.3 $\pm$ 3.6	84.6 $\pm$ 2.7	0.795 $\pm$ 0.029
CH-HNN	45.8 $\pm$ 3.9	63.7 $\pm$ 3.1	0.719 $\pm$ 0.035

B. Continual Learning Capability

The experiments in this subsection are organized around Table I, Table II, and Fig. 4. Table I follows the standard RL agent-sampling protocol, in which the learner collects training data by jointly sampling from the six available maps and is evaluated by averaging performance over the same evaluation maps. In contrast, Table II and Fig. 4 follow the continual-learning protocol described in Section III-A: the active sampler moves through the six maps sequentially, and after a map switch the environment no longer generates fresh trajectories from the previous map. Fig. 4 reports the average total-reward learning curves under this sequential protocol, while Table II reports the corresponding AP values after each map stage. The sequential protocol should therefore be interpreted as a diagnostic stress test for continual reinforcement learning, not as a generally better sampling strategy than standard joint multi-map data collection.

Comparing Table I with Table II first confirms an expected experimental tendency: policies trained under the standard RL agent-sampling protocol generally achieve higher final performance than policies evaluated under the sequential continual-learning stress test. With joint access to all six maps, the training distribution is closer to a stationary multi-map mixture, so the learner can repeatedly sample old and new scene regimes instead of relying on historical replay

after each domain switch. For example, under standard sampling, SAC reaches 67.3 $\pm$ 3.6 AP(DS), 84.6 $\pm$ 2.7 AP(RC), and 0.795 $\pm$ 0.029 AP(IS), whereas the same SAC-style learner fails to produce a valid policy by Town06 in the continual-learning sequence. SBA-UAS also performs better under standard sampling than under the sequential protocol, improving from the Town06 continual-learning endpoint of 60.2 $\pm$ 2.02 AP(DS), 77.5 $\pm$ 2.91 AP(RC), and 0.78 $\pm$ 0.013 AP(IS) to 76.85 $\pm$ 4.41, 90.16 $\pm$ 3.13, and 0.846 $\pm$ 0.018, respectively. More importantly, Table I shows that SBA-UAS exceeds PPO (Roach) under the standard protocol as well. This result is consistent with the central argument of this paper: RL algorithms are not necessarily weak because their basic optimization principles are inadequate; rather, large-scale multi-scenario autonomous-driving training can be constrained by loss of plasticity, catastrophic forgetting, and representation interference. The similarity-based activation, uncertainty-aware familiar-experience selection, and parameter-stabilization mechanisms mitigate these neural-network-level bottlenecks, so their benefit can appear not only in a strict continual-learning stream but also in ordinary joint multi-map RL training where heterogeneous driving regimes still compete for shared representational capacity.

TABLE II

AVERAGE PERFORMANCE (AP) OF ALL METHODS ACROSS THE CONTINUAL LEARNING SEQUENCE. VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER 10 RANDOM SEEDS; “-” DENOTES FAILURE TO PRODUCE A VALID POLICY.

Method	Map	AP(DS) $\uparrow$	AP(RC) $\uparrow$	AP(IS) $\uparrow$
PPO (Roach)	Town01	86.5$\pm$4.25	94.5$\pm$4.06	0.88 $\pm$ 0.036
	Town02	77.4 $\pm$ 2.93	91.2$\pm$5.28	0.80 $\pm$ 0.047
	Town03	8.2 $\pm$ 1.82	12.0 $\pm$ 2.28	0.60 $\pm$ 0.064
	Town04	1.4 $\pm$ 0.62	4.7 $\pm$ 2.88	0.57 $\pm$ 0.069
	Town05	-	-	-
	Town06	-	-	-
SAC	Town01	39.9 $\pm$ 4.90	40.2 $\pm$ 4.72	0.75 $\pm$ 0.041
	Town02	27.8 $\pm$ 3.49	36.5 $\pm$ 5.14	0.72 $\pm$ 0.025
	Town03	20.5 $\pm$ 2.64	31.1 $\pm$ 4.46	0.59 $\pm$ 0.043
	Town04	11.1 $\pm$ 2.13	20.4 $\pm$ 3.55	0.54 $\pm$ 0.042
	Town05	3.5 $\pm$ 2.35	7.9 $\pm$ 2.17	0.42 $\pm$ 0.056
	Town06	-	-	-
CH-HNN	Town01	50.4 $\pm$ 3.89	56.6 $\pm$ 3.34	0.78 $\pm$ 0.023
	Town02	39.7 $\pm$ 2.98	48.9 $\pm$ 2.07	0.76 $\pm$ 0.038
	Town03	34.0 $\pm$ 4.23	45.5 $\pm$ 3.61	0.71 $\pm$ 0.031
	Town04	32.5 $\pm$ 2.62	46.6 $\pm$ 3.75	0.69 $\pm$ 0.030
	Town05	31.7 $\pm$ 2.99	45.6 $\pm$ 4.39	0.66 $\pm$ 0.018
	Town06	31.1 $\pm$ 2.06	45.2 $\pm$ 4.63	0.65 $\pm$ 0.027
XdG	Town01	83.2 $\pm$ 4.54	90.5 $\pm$ 2.10	0.87 $\pm$ 0.025
	Town02	77.0 $\pm$ 2.15	88.1 $\pm$ 2.30	0.85 $\pm$ 0.024
	Town03	65.8 $\pm$ 4.46	83.4 $\pm$ 2.69	0.80 $\pm$ 0.033
	Town04	59.5 $\pm$ 3.31	75.2 $\pm$ 3.36	0.79 $\pm$ 0.028
	Town05	53.9 $\pm$ 2.71	70.5 $\pm$ 2.86	0.75 $\pm$ 0.019
	Town06	48.2 $\pm$ 3.77	64.3 $\pm$ 2.58	0.75 $\pm$ 0.025
SBA-UAS	Town01	82.5 $\pm$ 2.49	89.8 $\pm$ 1.70	0.89$\pm$0.029
	Town02	79.2$\pm$3.73	86.8 $\pm$ 2.29	0.87$\pm$0.022
	Town03	68.6$\pm$2.09	83.9$\pm$2.07	0.82$\pm$0.039
	Town04	66.7$\pm$2.64	81.5$\pm$2.85	0.81$\pm$0.026
	Town05	62.2$\pm$3.07	78.9$\pm$1.71	0.79$\pm$0.017
	Town06	60.2$\pm$2.02	77.5$\pm$2.91	0.78$\pm$0.013

Under the continual-learning protocol, Fig. 4 and Table II reveal clear differences in the ability of the compared methods to preserve competence over a non-stationary map stream. PPO (Roach) achieves strong early AP values on Town01

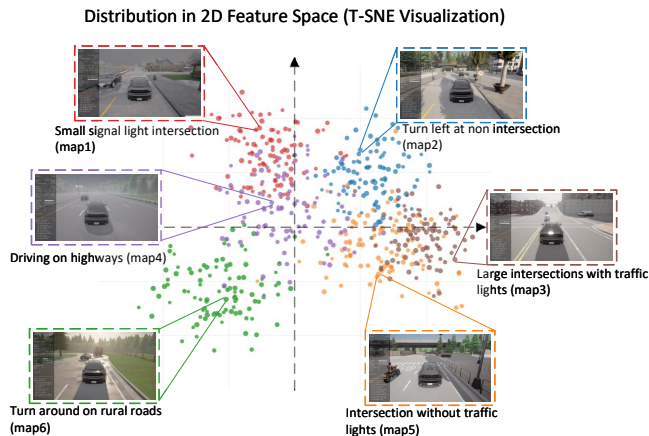


Fig. 5. We randomly sampled six groups of observations from six typical scenarios, with each group containing 60 samples. These observations were encoded by a ContextNet trained for 1.2×10^7 steps, and we then applied T-SNE to reduce the encoded outputs to a two-dimensional space for visualization. A representative example from each scenario type was selected, and the results show that ContextNet maps semantically similar scene observations into neighboring regions of the embedding space.

and Town02, but its on-policy sampling process loses direct access to old-domain trajectories after map switches; correspondingly, Table II shows a sharp degradation from Town03 onward and no valid policy after Town05. SAC benefits from off-policy replay at the beginning, yet its AP steadily decreases as later domains displace and interfere with earlier value representations, and it also fails by Town06. CH-HNN retains a complete six-map sequence, but its final AP(DS) of 31.1 ± 2.06 and AP(RC) of 45.2 ± 4.63 indicate a low performance ceiling when SNN-inspired memory mechanisms are coupled with gradient-based DRL. XdG maintains stronger continual-learning performance than these baselines, ending at 48.2 ± 3.77 AP(DS), but its map-level task partition remains coarse relative to the heterogeneous scenes inside each CARLA map. In contrast, SBA-UAS maintains the best AP values among task-agnostic continual-learning methods from the middle and later stages of the sequence and finishes at 60.2 ± 2.02 AP(DS), 77.5 ± 2.91 AP(RC), and 0.78 ± 0.013 AP(IS). The reward curves in Fig. 4 are consistent with this table-level pattern: SBA-UAS experiences transient drops at harder map transitions but preserves an overall recoverable and convergent trajectory. These results indicate that scene-conditioned pathway allocation and uncertainty-aware stabilization help the Critic retain useful old-domain value structure while still allowing adaptation to new driving domains.

Table III further quantifies the global performance of each method after the full sequence using three metrics: Average Performance (AP), Average Forgetting (AF), and Forward Transfer (FWT). PPO (Roach) and SAC (Large Buffer) are excluded from this comparison as they failed to produce any usable policy after learning all six maps. The aggregate scores indicate that SBA-UAS obtains a more favorable stability-plasticity tradeoff than the compared continual-learning baselines. In terms of AP, SBA-UAS achieves a DS of 60.22 ± 2.35 , compared with 48.20 ± 3.09 for XdG and 31.08 ± 3.38 for CH-

TABLE III
GLOBAL CONTINUAL LEARNING PERFORMANCE ACROSS METHODS IN TERMS OF AVERAGE PERFORMANCE (AP $\uparrow$), AVERAGE FORGETTING (AF $\downarrow$), AND FORWARD TRANSFER (FWT $\uparrow$). VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION UNDER THE SAME 10-SEED PROTOCOL USED FOR THE LEARNING-CURVE ANALYSIS.

Method	Metric	AP $\uparrow$	AF $\downarrow$	FWT $\uparrow$
CH-HNN	DS	31.08 ± 3.38	6.44 ± 1.74	10.08 ± 3.21
	RC	45.23 ± 2.26	6.14 ± 1.75	16.24 ± 2.44
	IS	0.65 ± 0.035	0.06 ± 0.010	0.32 ± 0.019
XdG	DS	48.20 ± 3.09	15.98 ± 1.69	26.12 ± 3.37
	RC	64.27 ± 2.79	14.18 ± 1.27	26.28 ± 2.30
	IS	0.75 ± 0.021	0.05 ± 0.009	0.51 ± 0.021
SBA-UAS	DS	60.22 ± 2.35	5.14 ± 0.81	26.94 ± 2.96
	RC	77.50 ± 1.73	2.06 ± 0.52	33.16 ± 1.74
	IS	0.78 ± 0.012	0.02 ± 0.006	0.54 ± 0.015

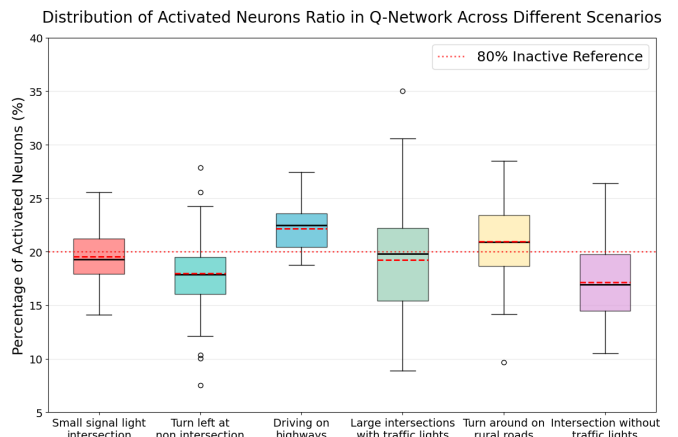


Fig. 6. Distribution of the proportion of activated neurons in the Q-network across six representative autonomous driving scenarios. The box plots summarize statistics from 60 observations per scenario: the solid line denotes the median, the dashed line the mean, the box spans the interquartile range (IQR), and the whiskers extend to the full data range. A horizontal red dashed line marks the 20% activation threshold.

HNN. For AF, which measures catastrophic forgetting, SBA-UAS records a DS forgetting rate of 5.14 ± 0.81 , lower than XdG's 15.98 ± 1.69 . This gap is consistent with a structural limitation of XdG: within a single CARLA map, heterogeneous subtasks (e.g., lane changing versus intersection navigation) coexist, whereas XdG uses coarse-grained, map-level task identifiers to partition its network. This mismatch between subnetwork allocation and intrinsic scenario structure limits its ability to isolate parameter interference. In contrast, SBA-UAS uses scene-conditioned pathway allocation and uncertainty-guided consolidation to implement finer-grained, task-ID-free isolation and reuse. The t-SNE visualization in Fig. 5 supports this interpretation by showing that functionally similar scenarios, such as intersections, are mapped to neighboring regions of the embedding space. On the FWT metric, SBA-UAS obtains the highest DS score (26.94), suggesting that the learned pathways reuse driving features shared across the tested CARLA domains, such as lane-keeping logic, and thereby support adaptation to later scenarios.

Fig. 6 illustrates the number of activated neurons in the

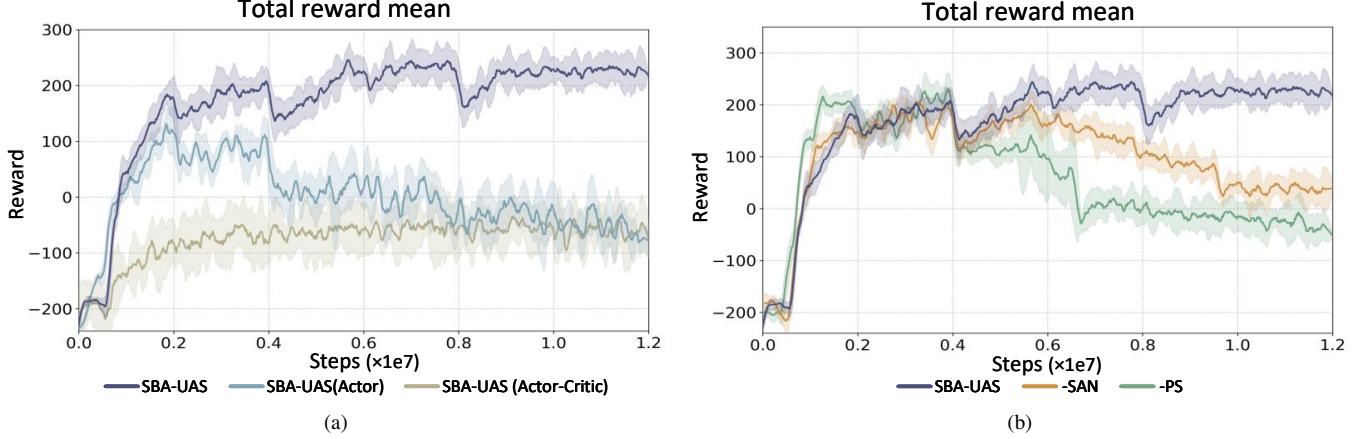


Fig. 7. Ablation analysis of architectural placement and Critic-side components. (a) Average reward convergence curves under different placements of the continual learning mechanism: **SBA-UAS (Actor)** applies the mechanism only to the Actor network, whereas **SBA-UAS (Actor-Critic)** incorporates it in both Actor and Critic networks. (b) Average reward convergence curves after removing the Similarity-Based Activation Network (-SAN) or disabling the Parameter Stabilization mechanism (-PS) in the Critic.

action-value (Q) network under identical autonomous driving observations. The results indicate that SBA-UAS activates scenario-specific pathways within the Q-network, leaving approximately 80% of parameters dormant in any given or similar scenario. This activation pattern provides empirical support for the sparse-pathway budget in Eq. (3): inactive capacity can later be recruited for new domains, while active pathways retain representations associated with previously encountered scenarios.

Despite these results, SBA-UAS—as an off-policy algorithm—remains susceptible to distributional shifts inherent in replay-based data usage. Existing studies suggest that uncertainty estimation can help diagnose such shifts [30], and the environment module in our architecture provides a basis for this direction. Future work will investigate how uncertainty-aware value estimation can further reduce distributional-shift effects during long-term continual learning.

C. Ablation Study

This section evaluates the design of SBA-UAS at two levels. The first level tests where the continual learning mechanism should be placed in the Actor-Critic architecture, and the second level tests which Critic-side components are responsible for the observed retention and adaptation behavior. Because the central claim concerns an unmodified Actor guided by a stabilized Critic, we further add diagnostic experiments that explicitly separate Actor behavioral degradation from Critic value-signal drift.

For the architectural-placement study, we construct two variants in addition to the proposed Critic-side SBA-UAS: (1) **SBA-UAS (Actor)**: inverting the proposed design, we equip the Actor with the SAN module and gating mechanism, while reverting the Critic to a standard CNN+MLP architecture (i.e., the base structure shown in Fig. 2(A)); (2) **SBA-UAS (Actor-Critic)**: both Actor and Critic are implemented as gated neural networks, sharing a unified activation signal generated by a single SAN module. As shown in Fig. 7(a),

TABLE IV
SUMMARY OF ABLATION AND DIAGNOSTIC RESULTS AFTER THE FULL SIX-MAP CONTINUAL-LEARNING SEQUENCE. VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER 10 RANDOM SEEDS.

Method	Metric	Value
SBA-UAS	AP(DS) $\uparrow$	60.20$\pm$2.71
	AF(DS) $\downarrow$	5.14$\pm$1.44
	FWT(DS) $\uparrow$	26.94$\pm$2.62
	$D_\pi \downarrow$	0.082$\pm$0.012
	$C_Q \uparrow$	0.886$\pm$0.025
SBA-UAS (Actor)	AP(DS) $\uparrow$	16.74 $\pm$ 4.59
	AF(DS) $\downarrow$	33.32 $\pm$ 5.73
	FWT(DS) $\uparrow$	0.65 $\pm$ 4.43
	$D_\pi \downarrow$	0.414 $\pm$ 0.049
	$C_Q \uparrow$	0.373 $\pm$ 0.049
SBA-UAS (Actor-Critic)	AP(DS) $\uparrow$	14.53 $\pm$ 3.89
	AF(DS) $\downarrow$	37.14 $\pm$ 7.77
	FWT(DS) $\uparrow$	-0.97 $\pm$ 3.89
	$D_\pi \downarrow$	0.474 $\pm$ 0.068
	$C_Q \uparrow$	0.321 $\pm$ 0.074
-SAN	AP(DS) $\uparrow$	36.20 $\pm$ 2.99
	AF(DS) $\downarrow$	18.55 $\pm$ 3.58
	FWT(DS) $\uparrow$	12.16 $\pm$ 3.06
	$D_\pi \downarrow$	0.266 $\pm$ 0.049
	$C_Q \uparrow$	0.568 $\pm$ 0.065
-PS	AP(DS) $\uparrow$	14.13 $\pm$ 2.96
	AF(DS) $\downarrow$	44.10 $\pm$ 5.63
	FWT(DS) $\uparrow$	7.73 $\pm$ 4.09
	$D_\pi \downarrow$	0.539 $\pm$ 0.054
	$C_Q \uparrow$	0.322 $\pm$ 0.050

these variants perform worse than the proposed method. **SBA-UAS (Actor)** begins with lower initial performance, loses learning capacity as early as the second map task, and reaches only 16.74 $\pm$ 4.59 AP(DS) after the full sequence. **SBA-UAS (Actor-Critic)** also suffers persistent convergence difficulty, obtaining 14.53 $\pm$ 3.89 AP(DS). These results indicate that merely adding memory-related structure to the Actor is insufficient when the Critic continues to provide unstable value targets, while simultaneously gating both networks increases the non-stationarity of the coupled policy-value optimization.

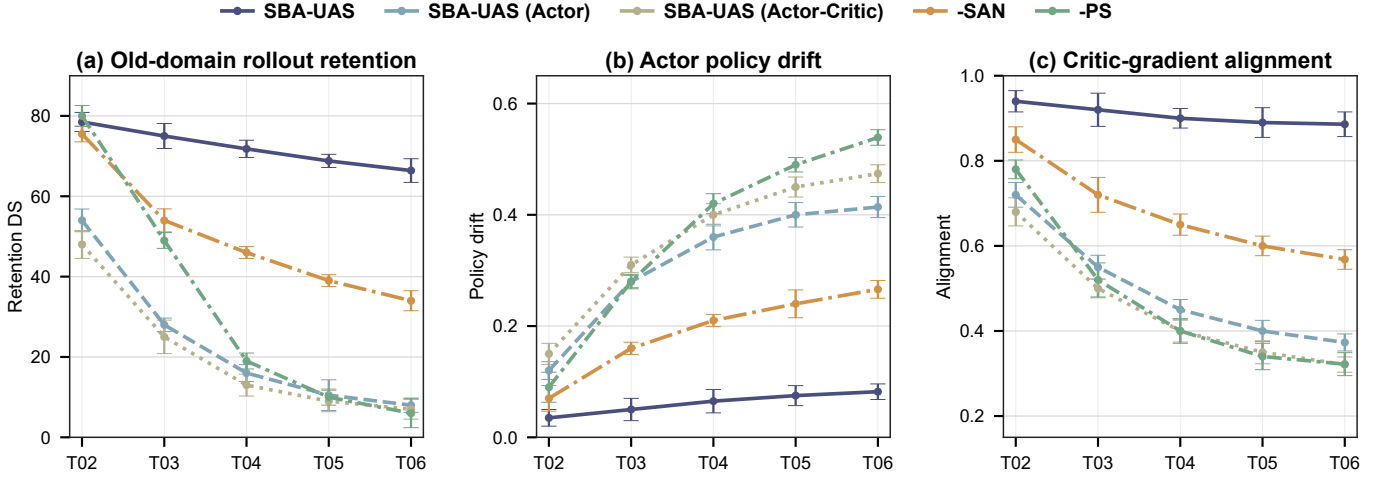


Fig. 8. Diagnostic analysis of Actor degradation under Critic-side continual learning. (a) Actor-only old-domain rollout retention evaluates the current Actor on previously learned maps with exploration disabled and without querying the Critic during execution. (b) Actor policy drift measures the divergence between the current Actor and historical high-performing Actors on saved old-domain states. (c) Critic-to-Actor supervisory-signal stability measures the alignment of value-gradient directions on historical states.

For the Critic-side component study, we use two additional variants: (1) **Removal of SAN (denoted as -SAN)**: this variant eliminates the Similarity-Based Activation Network (SAN), compelling the Q-network to rely on static, shared neural pathways for all scenarios, thereby preventing dynamic activation of specialized sub-modules based on observed states; (2) **Removal of Parameter Stabilization (denoted as -PS)**: this variant excludes the familiar experience buffer and the corresponding regularization constraints derived from uncertainty estimation, thus removing the protection mechanism for critical historical parameters during network updates. Fig. 7(b) and Table IV show that the -SAN variant remains close to the full SBA-UAS in structurally simple tasks such as Town01 and Town02, but its reward curve deteriorates after Town03, where greater environmental heterogeneity, including roundabouts and multi-lane intersections, increases representational interference. Its final AP(DS) decreases to 36.20 ± 2.99 and its AF(DS) rises to 18.55 ± 3.58 . Without context-aware pathway allocation, a monolithic Critic must encode heterogeneous features within shared capacity, making the value representation less robust under sequential domain shifts. The -PS variant achieves higher initial performance on Town01 and Town02 because unconstrained plasticity allows rapid current-task fitting, but it becomes vulnerable to catastrophic forgetting and policy degradation in later tasks, ending with 14.13 ± 2.96 AP(DS) and 44.10 ± 5.63 AF(DS). This pattern shows that SAN mainly mitigates representation interference, whereas Parameter Stabilization regulates the stability-plasticity tradeoff by protecting familiar-experience-related Critic parameters.

Although the preceding reward-level ablations support the Critic-side design choice, they do not by themselves isolate Actor degradation. We therefore perform an additional diagnostic experiment, summarized in Fig. 8. After each map is trained, the current Actor is evaluated on previously learned maps with exploration disabled and without invoking the Critic during action execution. This Actor-only rollout test directly

measures whether the executable policy has retained old-domain behavior. We also save historical observation sets $\mathcal{S}_i$ from each map and compute policy drift between the Actor checkpoint obtained after learning map i and the current Actor after learning map k :

$$D_\pi(i, k) = \mathbb{E}_{s \sim \mathcal{S}_i} [D_{\text{KL}}(\pi_{\theta_i}(\cdot|s) \parallel \pi_{\theta_k}(\cdot|s))]. \quad (24)$$

For the Critic-side supervisory signal, we further measure the alignment between historical and current value-gradient directions on the same old-domain states:

$$C_Q(i, k) = \mathbb{E}_{s \sim \mathcal{S}_i} \left[\cos(\nabla_a Q_{\phi_i}(s, a), \nabla_a Q_{\phi_k}(s, a))_{a=\mu_{\theta_k}(s)} \right]. \quad (25)$$

The diagnostic results show that SBA-UAS retains higher old-domain rollout performance, produces lower Actor policy drift ($D_\pi = 0.082 \pm 0.012$), and maintains higher Critic gradient alignment ($C_Q = 0.886 \pm 0.025$) than the variants without Critic-side stabilization. Thus, the unmodified Actor is not treated as intrinsically immune to forgetting. Rather, in the tested off-policy Actor-Critic setting, stabilizing the Critic preserves the value-gradient field on historical states, thereby reducing behaviorally harmful Actor drift and enabling stable policy evolution under sequential domain shifts.

In summary, the ablation results support a two-part interpretation. SAN supports adaptation to heterogeneous driving domains by reducing representation interference, and Parameter Stabilization protects familiar Critic knowledge from being overwritten. Together, these Critic-side mechanisms provide a more stable supervisory signal for the standard Actor, which explains why the proposed Critic-focused design outperforms Actor-only and Actor-Critic-gated variants in the tested continual CARLA stream.

VI. CONCLUSION

This work proposes the Similarity-Based Activation and Uncertainty-Aware Stabilization (SBA-UAS) architecture for

catastrophic forgetting and knowledge-transfer bottlenecks in deep reinforcement learning (DRL) for autonomous driving decision-making. By integrating selective feature sharing with uncertainty-driven parameter protection, SBA-UAS addresses the stability-plasticity dilemma in continual learning without requiring explicit task boundaries. Experimental results in the tested CARLA environment stream show improved historical-policy retention and adaptation to unseen maps and traffic conditions compared with the continual-learning baselines evaluated in this study. The core contributions of this work span three dimensions: (1) **Theoretically**, we present a DRL framework that jointly coordinates scene representation isolation and knowledge reuse, offering a computational model for continual reinforcement learning; (2) **Technically**, we introduce a task-id-free parameter stabilization paradigm grounded in predictive uncertainty quantification, thereby reducing the reliance on task identifiers that constrains conventional methods; (3) **Practically**, we provide a scalable architectural foundation for the continuous evolution of autonomous driving systems under sequential structured-domain changes.

Looking ahead, three limitations remain: the Bayesian environment model increases training-time memory-construction cost, although it is not required for deployment-time policy execution; the similarity mechanism depends on the quality of implicit representations; and the speed-scaled tests rely on normalized speed mapping, training-time braking attenuation in highway-like scenes, and PID execution, so they do not yet constitute full validation of higher-speed regimes with richer stochastic interactions or unstructured scenes. Future work will focus on lightweight environmental modeling, uncertainty-aware value estimation under distributional shifts, safety-aware memory retention, and broader real-world driving scenarios.

Code availability: Code, configurations, evaluation scripts, and checkpoints will be released at <https://github.com/thb-sau/SBA-UAS>.

REFERENCES

- [1] B. R. Kiran *et al.*, “Deep reinforcement learning for autonomous driving: A survey,” *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022.
- [2] L. Wang, X. Zhang, H. Su, and J. Zhu, “A comprehensive survey of continual learning: Theory, method and application,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5362–5383, Aug. 2024.
- [3] L. Gao, H. Tian, and J. Wu, “Efficient maximum entropy reinforcement learning framework guided by expert policies: An application in autonomous driving,” *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 15055–15067, Oct. 2025.
- [4] K. Yang, S. Li, Y. Chen, D. Cao, and X. Tang, “Towards safe decision-making for autonomous vehicles at unsignalized intersections,” *IEEE Trans. Veh. Technol.*, vol. 74, no. 3, pp. 3830–3842, Mar. 2025.
- [5] J. Wu, Z. Huang, W. Huang, and C. Lv, “Prioritized experience-based reinforcement learning with human guidance for autonomous driving,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 855–869, Jan. 2024.
- [6] N. Y. Masse, G. D. Grant, and D. J. Freedman, “Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization,” *Proc. Nat. Acad. Sci. U.S.A.*, vol. 115, no. 44, pp. E10467–E10475, Oct. 2018.
- [7] F. Zenke, B. Poole, and S. Ganguli, “Continual learning through synaptic intelligence,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 3987–3995.
- [8] J. Kirkpatrick *et al.*, “Overcoming catastrophic forgetting in neural networks,” *Proc. Nat. Acad. Sci. U.S.A.*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.
- [9] W. Huang and J. Parvizi, “Functionally diverse human insular architecture with memory-related hippocampal interactions,” *Nat. Neurosci.*, vol. 28, pp. 1580–1581, 2025.
- [10] T. Tadros, G. P. Krishnan, R. Ramyaa, and M. Bazhenov, “Sleep-like unsupervised replay reduces catastrophic forgetting in artificial neural networks,” *Nat. Commun.*, vol. 13, no. 1, Art. no. 7742, 2022.
- [11] S. Cho, H. Lee, S. Baek, and S. B. Paik, “Neuromimetic metaplasticity for adaptive continual learning without catastrophic forgetting,” *Neural Netw.*, Art. no. 107762, 2025.
- [12] B. J. Gutiérrez, Y. Shu, W. Qi, S. Zhou, and Y. Su, “From rag to memory: Non-parametric continual learning for large language models,” *arXiv preprint arXiv:2502.14802*, 2025.
- [13] K. Deja, B. Cywiński, J. Rybarczyk, and T. Trzciński, “Adapt & align: Continual learning with generative models’ latent space alignment,” *Neurocomputing*, Art. no. 130748, 2025.
- [14] D. W. Zhou *et al.*, “Dual consolidation for pre-trained model-based domain-incremental learning,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 20547–20557.
- [15] H. Zhuang *et al.*, “Online analytic exemplar-free continual learning with large models for imbalanced autonomous driving task,” *IEEE Trans. Veh. Technol.*, 2024.
- [16] Y. Meng *et al.*, “Preserving and combining knowledge in robotic lifelong reinforcement learning,” *Nat. Mach. Intell.*, pp. 1–14, 2025.
- [17] S. Dohare, J. F. Hernandez-Garcia, Q. Lan, P. Rahman, A. R. Mahmood, and R. S. Sutton, “Loss of plasticity in deep continual learning,” *Nature*, vol. 632, no. 8026, pp. 768–774, 2024.
- [18] M. Hattori, “A biologically inspired dual-network memory model for reduction of catastrophic forgetting,” *Neurocomputing*, vol. 134, pp. 262–268, 2014.
- [19] R. Wu *et al.*, “Information-theoretic dual memory system for continual learning,” *Knowl.-Based Syst.*, Art. no. 113913, 2025.
- [20] Q. Shi *et al.*, “Hybrid neural networks for continual learning inspired by corticohippocampal circuits,” *Nat. Commun.*, vol. 16, no. 1, Art. no. 1272, 2025.
- [21] M. T. Spaan, “Partially observable Markov decision processes,” in *Reinforcement Learning: State-of-the-Art*, Berlin, Heidelberg: Springer, 2012, pp. 387–414.
- [22] Z. Zhang, A. Liniger, D. Dai, F. Yu, and L. Van Gool, “End-to-end urban driving by imitating a reinforcement learning coach,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 15222–15232.
- [23] X. Jia *et al.*, “DriveAdapter: Breaking the coupling barrier of perception and planning in end-to-end autonomous driving,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 7953–7963.
- [24] J. Hwangbo *et al.*, “Learning agile and dynamic motor skills for legged robots,” *Sci. Robot.*, vol. 4, no. 26, Art. no. eaa5872, 2019.
- [25] L. V. Jospin, H. Laga, F. Boussaid, W. Buntine, and M. Bennamoun, “Hands-on Bayesian neural networks—A tutorial for deep learning users,” *IEEE Comput. Intell. Mag.*, vol. 17, no. 2, pp. 29–48, May 2022.
- [26] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [27] H. Xu, L. Szymanski, and B. McCane, “VASE: Variational assorted surprise exploration for reinforcement learning,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1243–1252, Mar. 2023.
- [28] CARLA Team. (2020). *CARLA autonomous driving leaderboard*. [Online]. Available: <https://leaderboard.carla.org/>
- [29] T. Haarnoja *et al.*, “Soft actor-critic algorithms and applications,” *arXiv preprint arXiv:1812.05905*, 2018.
- [30] C. Bai *et al.*, “Pessimistic bootstrapping for uncertainty-driven offline reinforcement learning,” *arXiv preprint arXiv:2202.11566*, 2022.
- [31] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, “Curriculum learning,” in *Proc. 26th Int. Conf. Mach. Learn. (ICML)*, Montreal, QC, Canada, 2009, pp. 41–48.
- [32] T. Matisen, A. Oliver, T. Cohen, and J. Schulman, “Teacher-student curriculum learning,” *arXiv preprint arXiv:1707.00183*, 2017.
- [33] P.-W. Chou, D. Maturana, and S. Scherer, “Improving stochastic policy gradients in continuous control with deep reinforcement learning using the Beta distribution,” in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 70, 2017, pp. 834–843.